

Coding & Solution

Date	15 July 2026
Project Name	Smart Lender – AI Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub - akshaya2006-13/Smartlender · GitHub
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy
Key Features Implemented	• Loan eligibility prediction using Machine Learning. • Data preprocessing and feature engineering. • Training using Decision Tree, Random Forest, KNN, and XGBoost algorithms. • Selection of the best-performing model (XGBoost). • Web-based interface for loan prediction. • Instant prediction based on applicant details. • IBM Cloud deployment support.
Pending / Incomplete Features	User authentication, prediction history database, email notifications, admin dashboard, report generation
Setup / Run Instructions	1. Clone the repository. 2. Install Python 3.8+. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Run the application using <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000/</code> in a web browser. 6. Enter applicant details and click Predict to view the loan approval result.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	No

Additional Notes / Comments:

The Smart Lender project is developed using Python, Flask, and Machine Learning libraries to automate loan approval prediction. The application follows a modular structure with separate components for data preprocessing, model training, evaluation, and prediction. XGBoost was selected as the final model after comparing multiple classification algorithms due to its superior performance. The source code is maintained in a GitHub repository to support collaboration, version control, and future enhancements.